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Module Handbook

OCOM5203M Deep Learning D02 (23/24)

Contents

Module Handbook 1
Contents 1
Accessibility 2
Welcome to the Module 3
Module Team 3
Module Objectives 3
Learning Outcomes 3
Syllabus 5
Unit 1 – Optimisation and Pytorch 5
Unit 2 – Deep feedforward networks 5
Unit 3 – Convolutional neural networks 5
Unit 4 – Sequential data 6
Unit 5 – Relational data 6
Unit 6 – Deep generative models 6
Key Dates 7
Module Assessment 8
Assessment 1: On-line test 8
Assessment 2: Practical assignment on image caption generation 11
Reading List 14

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Additional Information and Support 14

Key Documents 14

Skills@Library 14

General Module Forum 14

Student Success Advisor 14

Recording Policy 15

Complaints 15

Accessibility

For information in alternative formats please email DESstudents@leeds.ac.uk or speak with your Student Success Advisor.

Welcome to the Module

Welcome to OCOM5203M Deep Learning.

This handbook contains administrative information related to the module to help you to organise study time for the module. We hope that you find this material useful and the work you do on the module rewarding.

Module Team

Your module leader is Nabi Omidvar. You will have the opportunity to speak to Nabi during the live weekly webinar sessions. If you need to contact the module leader directly, you can find their details in the 'Staff Information' section of Minerva. Nabi is a University Academic Fellow with a particular interest in optimisation and the application of machine learning in the finance sector.

Module Objectives

This module is designed to introduce you to the field of Artificial Intelligence (AI), taking a strongly integrative and state of the art approach based on deep neural networks. In line with the use of AI in key sectors (e.g. finance, health, law), there is an emphasis on the combination of multiple input modalities – specifically, combining images, text and structured data. Students gain hands-on experience in developing AI systems to address real-world problems, providing the knowledge and skills necessary to develop an AI system as part of an MSc project.

Learning Outcomes

By the end of this module students will be able to:

- Design and implement AI systems in a language such as Python, based on machine learning
- Apply deep learning in standard AI tasks (e.g. image classification, sentiment analysis)

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- Represent data from various modalities (e.g. text, images, records) for multi-modal deep learning (e.g. translating images into textual captions, translating text into cognitive categories such as human sentiment)
- Critically evaluate systems, using standard performance metrics
- Demonstrate a detailed understanding of the technical aspects of deep learning, including the different kinds of architecture, limitations, dependence on data and computational requirements
- Apply your knowledge to address challenges from a specific sector; for example, in detecting financial fraud from transaction data, and in informing medical diagnosis through combining medical images and structured data from electronic patient records.

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Syllabus

Unit 1 – Optimisation and Pytorch

You will learn some preliminary concepts such as manipulating image data using Python and Jupyter Notebook, function optimisation which is fundamental to machine learning, and the fundamental principles of PyTorch to implement a simple optimisation algorithm called gradient descent. You will then learn about what deep learning is and how it relates to artificial intelligence and machine learning. You finally explore linear classifiers and a classification algorithm called logistic regression to motivate the key ideas in deep learning.

Unit 2 – Deep feedforward networks

You will learn about a type of neural network, multilayer perceptron, and how it can be constructed by stacking up the building blocks you learned about in the previous unit. You will explore the backpropagation algorithm to train a neural network using data and construct your own networks in PyTorch and evaluate their performance on classification problems. You will produce diagrammatic representations of network architectures and be able to explain the concept of momentum used in optimisation.

Unit 3 – Convolutional neural networks

You'll explore how to describe the use of convolution and the link to natural vision systems. You'll learn about the role of convolution in the detection of visual features and the idea of encoding an image in terms of the dominant features. You'll apply convolution to an image and produce masks tailored to finding specific features. You will be able to specify and explain the architecture and components of a convolutional neural network (CNN), explain various methods used for regularisation of training a CNN and communicate the notion and significance of a receptive field. You'll implement a CNN for image classification in PyTorch and monitor training performance on test data. You'll be able to explain both the idea of semantic segmentation and the form of training data and how the performance of a segmentation method can be evaluated.

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Unit 4 – Sequential data

You will look at a range of tasks from different domains that involve sequential data and be able to explain the mathematical concept of a stochastic process. You'll explore the principle of using a 1D convolutional neural network to process sequential data and be able to describe one of the main ways for representing individual words (or other tokens) in an embedding space. You will learn to describe the architecture of a recurrent neural network (RNN) and how it is used and trained. You'll be able to explain the application of an RNN to text generation. You will learn about two developments of the vanilla RNN that provide a long-term memory: LSTMs and GRUs. You will learn about the recently developed transformer models that are now being used to produce stochastic models of language from very large datasets of on-line text.

Unit 5 – Relational data

You will explore how relational data can be represented as a graph structure and be able to explain the use of relational data in representation and reasoning. You'll learn about the structure of a graph neural network (GNN) and the way in which inference occurs. You'll explore the use of PyTorch Geometric for building GNNs and be able to describe contemporary applications of a GNN.

Unit 6 – Deep generative models

You will look at the architecture and training methodology of a Generative Adversarial Network (GAN) and be able to describe a well-known application of a GAN for the generation of images. You will use this knowledge to examine applications of generative models in image-to-image translation.

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Key Dates

	Monday	Tuesday	Wednesday	Thursday	Friday
Unit 1	**30 Oct** **Module starts**	**31 Oct**	**1**	**2**	**3**
Unit 2	**6** **Unit 1 Webinar**	**7**	**8**	**9**	**10**
Unit 3	**13** **Unit 2 Webinar**	**14**	**15**	**16**	**17**
Unit 4	**20** **Unit 3 Webinar**	**21**	**22**	**23**	**24**
Unit 5	**27** **Unit 4 Webinar** **Assessment 1 Release**	**28**	**29**	**30** **Assessment 2 release**	**1**
Unit 6	**4** **Unit 5 Webinar** **Assessment 1 deadline**	**5**	**6**	**7**	**8**
Unit 7	**11** **Unit 6 webinar with Q&A**	**12**	**13**	**14**	**15**
Study break	**18** **Assessment 2 deadline**	**19**	**20**	**21**	**22** **University closed**

Please see the module area in Minerva for specific timings for webinars and assessment submission deadlines.

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Module Assessment

You should familiarise yourself with the Digital Education Service Code of Practice on Assessment (CoPA) before you start your first assignment. The CoPA contains important information about marking, late submissions, resits, mitigating circumstances and more.

Assessment 1: On-line test

Summary

An online test relating to the topics covered in the module.

Deadline

Due: Monday 4th December at 08:30am UTC (London)

Purpose

The purpose of this assignment is to test your understanding of the various topics covered in the module. It is not a test of memory, and you may consult any of the material associated with the module in answering the questions.

Task

This is an online test with multiple questions relating to any of the topics covered in the module, including the academic papers that you are asked to read in certain units.

Once you begin you will have 6 consecutive hours to complete the test, but it is not expected that you will need to spend more than 2 to 3 hours in total. You will choose when to start this period of 6 hours and you must ensure that you submit your assignment by the end of this period.

The test will be available from Monday 27th November at 08:30am UTC (London).

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Academic integrity

As a student at the University of Leeds, you need to adhere to the University's regulations on academic misconduct. All work you submit for assessment is expected to be your own independent work. Where you have worked with others or where you present other people's work as part of your assessment, you must fully acknowledge this.

All students should ensure that they are aware of the principles of Academic Integrity. For more information or if you are unsure about academic integrity, please read the University of Leeds' guidance on Academic Integrity. Please note that by submitting this piece of work you are agreeing to the University's Declaration of Academic Integrity.

Submission

Assessment 1 should be submitted via Gradescope.

The link to access the online test assignment will be available via Submit my Work once the assessment window is open.

The assignment will be available for one week. In that week, you have 6 hours in which to complete and submit your assignment. For example, if you begin the assessment at 10am on Friday 1st December, you must complete it by 4pm that day. Please note the time will begin as soon as the assessment is accessed, so only start when you are ready.

Your answers for this assignment will be automatically saved as you go through and once you have completed you will need to submit them.

The timer will continue counting down even if you leave the Gradescope site.

Once you have submitted, you will see a review screen where the questions and your given answers will appear on the left hand side. The points value that appear against each question are the marks available, and do not reflect your actual score.

For this assignment type, you will not receive email confirmation of your submission.

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If you have partially completed the assessment by the deadline, your assessment will be automatically submitted and marked.

Marking

This is a compulsory assessment for OCOM5203M Deep Learning. It is worth 25% of your overall module grade.

Feedback

Your grade will be visible via 'Gradebook' on Minerva once marking has been completed. We will notify you by email when your grade is available, usually within 15 working days of the assignment deadline.

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Assessment 2: Practical assignment on image caption generation

Summary

Build a system for generating textual captions from images and evaluate the performance.

Deadline

This assignment will be released on Thursday 30th November at 08:30am UTC (London). The notebook file for the assignment will be made available on the 30th in the Assessment 2 folder in Minerva.

Due: Monday 18th December 2023 at 08:30am UTC (London)

Purpose

The purpose of this assignment is to develop skills in building and evaluating neural networks that involve both images and text, including text pre-processing, performance metrics and alternative architectures.

Task

Develop code for textual caption generation from an image. The code and results of experiments are contained within a Jupyter notebook.

Academic integrity

As a student at the University of Leeds, you need to adhere to the University's regulations on academic misconduct. All work you submit for assessment is expected to be your own independent work. Where you have worked with others or where you present other people's work as part of your assessment, you must fully acknowledge this.

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All students should ensure that they are aware of the principles of Academic Integrity. For more information or if you are unsure about academic integrity, please read the University of Leeds' guidance on Academic Integrity. Please note that by submitting this piece of work you are agreeing to the University's Declaration of Academic Integrity.

Submission

Submit your work by the due date and time via Minerva using the 'Submit my work' link in Minerva. Please submit the following:

Please submit the following:

1. Completed Jupyter notebook file, in .ipynb format. **Do not change the file name or the automatic grading will be affected.**

2. The .html version of your notebook; File > Download as > HTML (.html). Check that all cells have been run and all outputs (including all graphs you would like to be marked) displayed in the .html for marking.

Please note: You may need to upload these to Minerva as a zip file if submission of single files isn't possible.

Please be aware that once you have uploaded your file/files, you will need to click on submit button. Failure to do so will result in your submission not being logged with Minerva.

Furthermore, if you are planning to submit late, or have an extension, please note that you should only submit to the late submissions area. You should not submit a draft to the initial submission link as a submission may be logged for you and the Digital Education Service will only accept the coursework which is submitted by the deadline, regardless of whether it is a draft submission.

Marking

This is a compulsory assessment for OCOM5203M Deep Learning. It is worth 75% of your overall module grade.

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Feedback

Concise comments and marks relating to individual sections of the assignment.

Your grade will be visible via 'Gradebook' on Minerva once marking has been completed. We will notify you by email when your grade is available, usually within 15 working days of the assignment deadline.

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Reading List

The reading list and online course readings for this module can be found in the 'Reading List' section of Minerva.

Additional Information and Support

Key Documents

Please ensure that you read and familiarise yourself with the following important documents, available in the Sustainable Business Leadership Online organisation area on Minerva:

- Digital Education Service (DES) Code of Practice on Assessment
- Code of Practice on Attendance

Skills@Library

Skills@Library provides support for students to develop academic skills. This includes areas such as researching, writing, note making and referencing. This support is delivered through online resources, workshops and webinars: http://library.leeds.ac.uk/skills.

General Module Forum

The Digital Education Service and the University of Leeds are committed to supporting students. If you have any questions about the academic content of the module, please post them in the module's 'General Module Forum' discussion board. You can access this through the 'Discussion Board' section of Minerva. The Module Team will reply to your post and your peers will also have access to the answer.

Student Success Advisor

If you have any questions about using Minerva, administrative processes, or pastoral care please contact your Student Success Advisor. They will be able to advise you or signpost you to the correct resource.

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Recording Policy

As part of your study on this module, webinars will be recorded by the University. This recording is conducted in accordance with the University's Policy on Audio or Video Recording for Educational Purposes. Please read the Policy for further background information relating to your rights. The purpose of the recording is to support your study at the University. It will be accessible by students enrolled on the module through Minerva.

In accordance with the University's Intellectual Property Policy in general where staff or students make a recording, all rights in that recording (including rights in the sounds made) will belong to the University. The University seeks to respect student and staff rights to be acknowledged as authors and performers. Otherwise, to the extent allowed under the general law, any individual interests in the recordings are waived, allowing the University to act in effect as a custodian, to maximise educational objectives in the general interest of all.

If you have any concerns relating to the recording please contact the individual leading the educational activity, or the module leader. If you wish to opt out of being recorded or wish to request that your contribution be edited please contact the individual leading the activity or the module leader in advance of the session (for optouts) or immediately after the session (for requests for edits). Please note that you may not opt out of recordings which form part of the assessment process on the module. If you have any concerns relation to the recording, please contact your Student Success Advisor.

Complaints

If you have any complaints about this module, please write to DESStudents@leeds.ac.uk and the Digital Education Service will review your complaint. If a remedy cannot be arrived at, the University's official complaints procedure could be followed.

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